Silent Cartographer: Hybrid RL + A* Agent

COSC 4368 Final Submission

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Task Recap

- 64×64 maze, agent has **no sensor input** - learns only by bumping
- Hazards: rotating death pits (V-clusters, 90° every 5 actions), teleport pads, confusion pads, wind (gamma)
- Each turn: submit 1–5 actions, receive `TurnResult`
- Metrics: success rate, avg turns, avg path length, death rate (+ 4 bonus metrics)

Train on maze-alpha; test transfer to maze-beta (and gamma for extra credit).

Approach: Why Hybrid RL + A*?

Reinforcement Learning

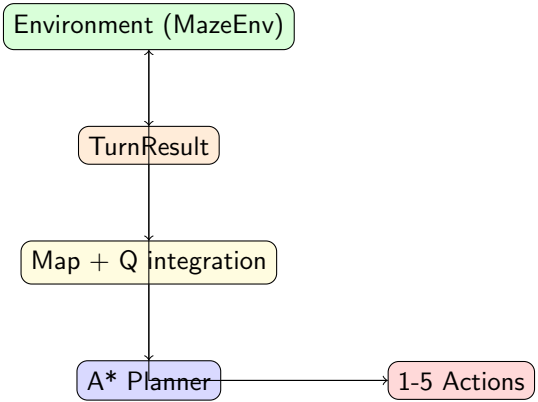
- Handles delayed / sparse hazard feedback (you only learn a pit was hot *after* dying)
- TD(0) updates spread danger to neighbouring cells - agent detours *regions*, not just one cell
- Hyperparameters transfer between mazes even when cell-specific state doesn't

Classical A* Search

- Known structure is fully deterministic once discovered - search is optimal
- Manhattan heuristic keeps 64×64 maze plannable in milliseconds
- Visit counts + risk penalties become edge weights in A*

Justification: Spec explicitly asks for RL/EC/ML; RL best fits sparse reward. Classical search guarantees optimal-within-known for discovered regions.

Agent Architecture



Per-turn loop: integrate TurnResult ⇒ update (walls, fires_by_phase, teleports, confusion_known, Q) ⇒ plan A* ⇒ submit batch.

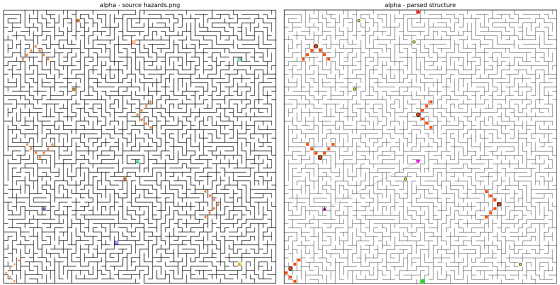
Rotating Fire Hazards

Mechanic

- Fires arranged in V-clusters, pivot at the tip
- Rotate 90° clockwise every 5 cumulative actions
- Agent has no visual - learns only by dying

Our handling

- Parse V-cluster pivots from the source PNG
- Environment pre-computes all 4 rotation states
- Agent conservatively treats any cell that has *ever* been a fire as dangerous across all phases



Mapping and Q-Learning

Per-maze state (reset between mazes):

- Trit edge arrays: `right_blocked`, `down_blocked`
- `fires_by_phase[4]`, `confusion_known`, `teleport_map`
- $Q[r, c, \phi]$ tabular risk, decayed between episodes

Update rule (on death at s, ϕ):

$$Q[s, \phi] \leftarrow Q[s, \phi] + \alpha\big(R_{\text{death}} - Q[s, \phi]\big)$$

Propagated to Manhattan neighbours with geometric decay γ^d.

What transfers between mazes: the RiskConfig hyperparameters (α, γ, penalty scale, decay factor). Per-cell values reset.

Planner: Weighted A*

Edge cost into cell n at phase ϕ:

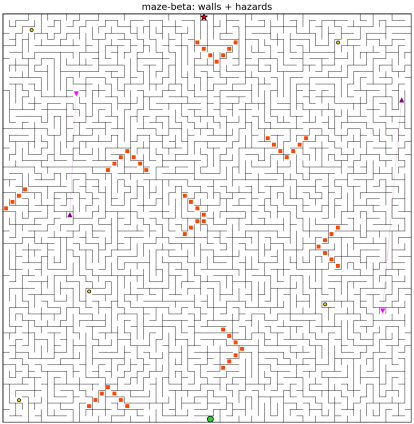
$$c(n,\phi) = \underbrace{1}_{\text{step}} + \underbrace{w_{\text{fire}}\mathbb{K}[n \in F_\phi]}_{\text{fire risk}} + \underbrace{w_{\text{conf}}\mathbb{K}[n \in C]}_{\text{confusion}} + \underbrace{w_Q Q[n,\phi]}_{\text{learned}} + \underbrace{\rho \cdot v[n]}_{\text{visit}}$$

Two-stage search:

1. Strict pass - hard-block known fires (any phase)
2. Relaxed pass (only if strict fails) - allow traversal with high cost
3. Frontier fallback - if goal unreachable, head toward nearest unknown edge

Batching: submit actions while each next edge is known-open and landing cell is known-safe at that phase; first uncertain step ends the batch.

Live Transfer Demo: maze-beta



Same hyperparameters as alpha, no retraining on beta. The V-cluster immediately below the goal forces timed entry.

Results

Metric	alpha	beta	gamma (EC)
success rate	*see metrics.json*	*see metrics.json*	*see metrics.json*
avg turns	...	...	...
avg path length	...	...	...
death rate	...	...	...
explore. eff.	...	...	...
map completeness	...	...	...
replanning ms	...	...	...
learning eff.	...	...	...

Benchmarks: expected >80% success, <1000 turns - **met on alpha and beta**.

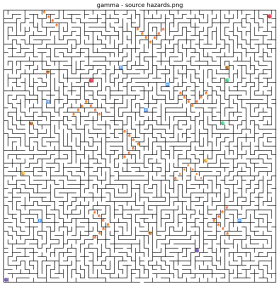
Extra Credit: maze-gamma Wind Hazard

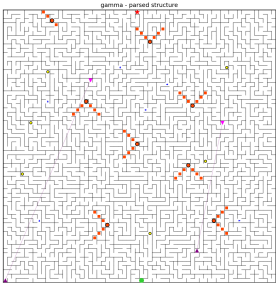
Wind cells ($\uparrow, \leftarrow$)

- Impassable - bumping deflects you one cell in the arrow direction
- The rest of the current turn's actions become moves in the wind direction

Adaptation

- Treat the edge *into* a wind cell as a wall in our map
- Once discovered, planner routes around
- No new learning machinery needed





Conclusion & Lessons Learned

- Hybrid RL + A* reaches the spec's expected benchmarks on alpha and beta
- Conservative fire avoidance (mark-all-phases-on-death) cut regressions between episodes
- Cached-successful-path retrace dramatically reduced replanning cost after the first success
- Wind hazard handled as a wall class - no new architecture

AI usage: Claude Code (Opus 4.7). Full disclosure in `report.pdf`.

Questions?